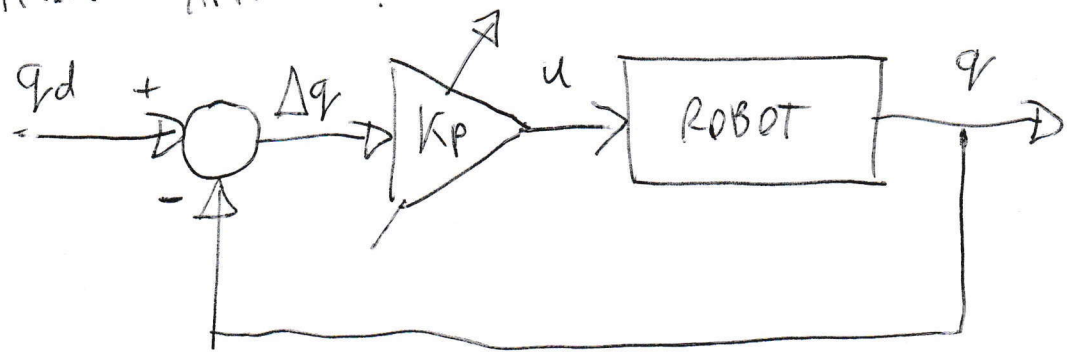
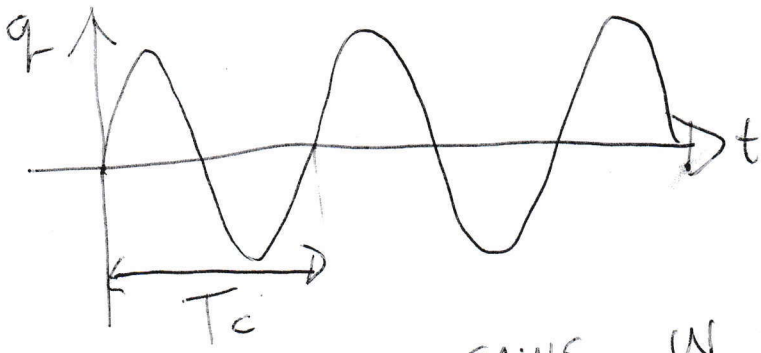
ZIEGLER NICHOLS TUNING FOR PID.

OSCILLATION APPROACH.

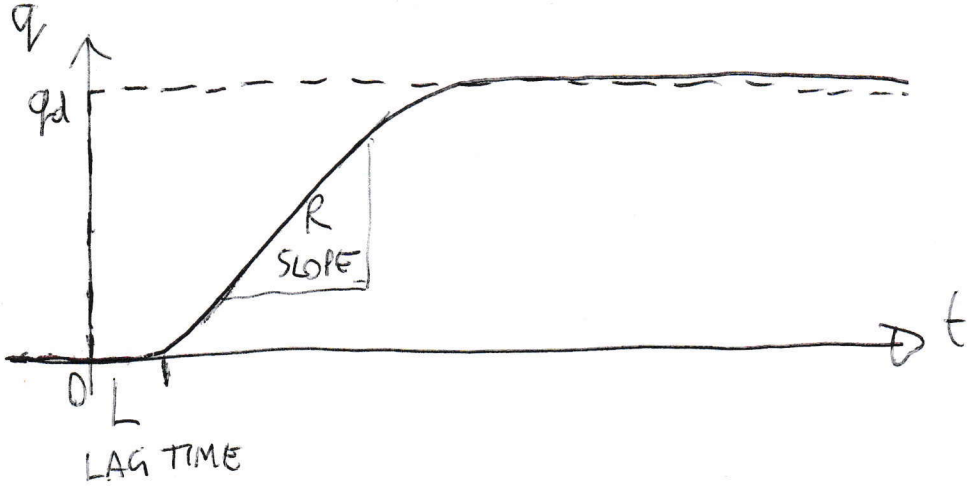


INCREASE GAIN K_p UNTIL IT REACHES K_c



SUBSTITUTE INTO GAINS IN Z-N TABLE.

OPEN - LOOP STEP APPROACH.



PID "3 PROBLEM CHILDREN"

PROPORTIONAL P – "FAST CHILD" – CAUSE OVERSHOOTS OR OSCILLATIONS IF GAIN TOO LARGE.

INTEGRAL I – "GREEDY CHILD" – WILL USE UP ALL AVAILABLE BANDWIDTH IF NOT KEPT IN CHECK (INTEGRAL ANTI-WINDUP).

DERIVATIVE D – "NOISY CHILD" – WILL RESPOND TO NOISE IN THE SYSTEM IN AN ADVERSE MANNER. (FILTER NOISE).